

UNIT 2: Data Centre Electrical Infrastructure

Easy-to-Understand Notes with Real-World Analogies, Solved Numericals & Ready Exam Answers

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1. The Core Idea: Why is Electrical Design so Critical?

Think of a Data Centre as a **giant factory whose only raw material is electricity** and whose finished product is digital data.

Analogy: The Hospital ICU

If power cuts in an apartment building, the TV shuts off and you light a candle. But in an ICU, patients die if oxygen stops for 10 seconds. Similarly, in a Tier III/IV data centre, if power flickers for even **15 milliseconds**, thousands of servers reboot, banking transactions fail, and millions of dollars are lost. The electrical system must have **zero single points of failure**.

⚡ 2. Utility Power Systems & Transformers

A. Dual Utility Feeds

Never take power from one street pole. A Tier III/IV data centre connects to **two separate electrical substations** fed by different transmission grids.

Exam Keyword: Physical Routing Diversity

The two underground cable trenches must enter the property boundary from opposite directions (at least 20 meters apart). If a construction backhoe cuts one cable, the other feed remains 100% intact.

B. Cast Resin Dry-Type (CRT) vs. Oil Transformers

Parameter	Cast Resin Dry-Type (CRT)	Oil-Cooled Transformer
Cooling Medium	Air flowing over epoxy-resin coils	Flammable mineral oil
Fire Danger	Zero Fire Risk (Self-extinguishing)	High Fire Risk (Oil explodes/burns)
Indoor Installation	YES (Directly beside IT rooms)	NO (Must be outside with blast walls)
Data Centre Standard	Universal Industry Standard	Used only in outdoor utility yards

C. The K-Factor: Why Computer Servers Murder Transformers

Analogy: Smooth Driving vs. Potholes

A normal light bulb draws electricity smoothly like driving on a smooth highway. A server power supply (SMPS) takes choppy, violent bites of current thousands of times a second (Non-linear load). This creates electrical shockwaves called **Harmonics**.

Harmonics cause high-frequency Eddy current heating inside transformers proportional to the **square of the frequency**. Normal transformers catch fire. We must use **K-Factor Rated Transformers** (specifically **K-20** for data centres), which have thicker copper coils, electrostatic shields, and oversized neutrals.

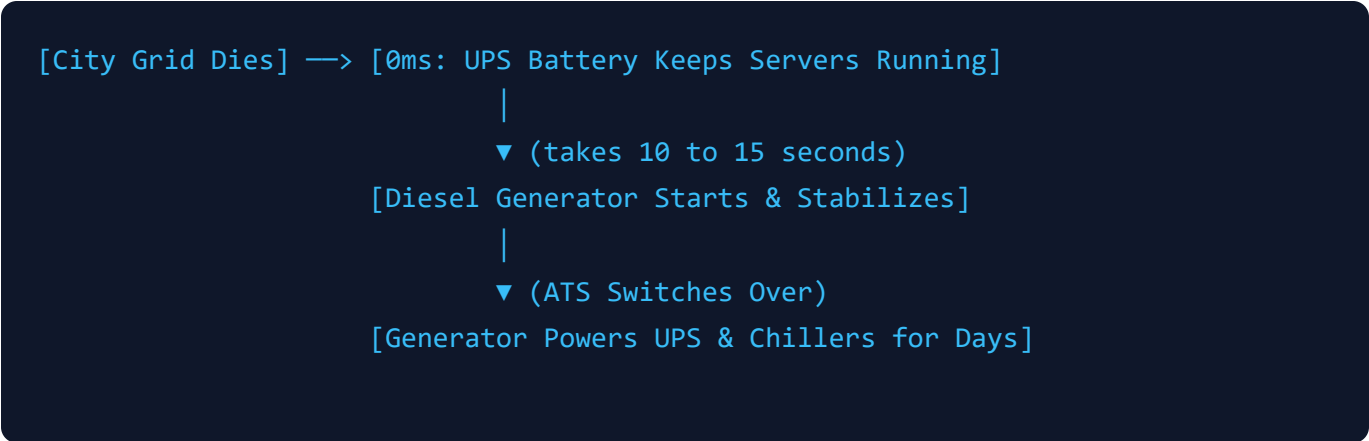
D. Triplen Harmonics & 200% Neutral

3rd, 9th, and 15th harmonics do not cancel out in the neutral wire—they **add up**! Neutral current can reach **1.73 times the phase current**. Therefore, data centre cables use a **200% double-sized neutral conductor** to prevent melting.



3. Backup Power: Diesel Generators & UPS Topologies

A. The Lifesaving Handshake



B. Generator DCC Rating (Data Centre Continuous)

Normal building backup generators have an **ESP (Emergency Standby Power)** rating (limited to 200 hours/yr and 70% load). **The Uptime Institute rejects ESP!** Data centres require **DCC (Data Centre Continuous)** rated generators that can run at 100% full capacity for unlimited days, with at least **48 to 72 hours of on-site diesel storage**.

C. Online Double Conversion UPS (VFI)

Why is it the only UPS used in Tier III/IV?

- **Double Conversion:** AC (utility) —> DC (clean power) —> AC (brand new pure sine wave).
- **Zero Transfer Time (0.00 ms):** The battery sits permanently on the internal DC bus. When grid power drops, the battery provides power instantly with no mechanical relay click.

D. VRLA vs. Lithium-Ion Batteries

Feature	VRLA (Lead-Acid)	Lithium-Ion (LiFePO4)
Weight & Size	Very heavy (needs floor reinforcement)	60% lighter & 50% smaller
Lifespan	3 to 5 years	12 to 15 years (lasts whole UPS life)

Room Cooling	Strictly 20°C–25°C (heat kills lead acid)	Works comfortably at 30°C (saves AC power)
Recharge Speed	12 to 24 hours	1 to 2 hours fast recharge



4. Power Distribution: ATS vs. STS & Busways

⚠️ Classic Exam Trap: Never confuse ATS and STS!

- **ATS (Automatic Transfer Switch):** Mechanical switch. Takes **50 to 100 milliseconds**. Used upstream between City Grid and Diesel Generator.
- **STS (Static Transfer Switch):** Solid-state silicon SCR switch. Takes only **2 to 4 milliseconds** (1/4 of an AC cycle). Used downstream at server racks so servers never lose power.

Overhead Track Busway vs. Floor Whips

Old data centres ran hundreds of thick copper cables ("whips") under raised floor tiles, blocking air cooling. Modern data centres hang **aluminum track busways from the ceiling**. Racks are powered via simple twist-and-lock "Tap-off boxes" with zero floor blockage.

5. Earthing, Grounding & Lightning Protection

A. The TN-S Earthing Scheme (Mandatory for IT)

T (Transformer neutral connected to earth), **N** (Equipment chassis bonded), **S** (Neutral and Protective Earth wires are kept **STRICTLY SEPARATE** throughout the building).

Why TN-C is BANNED: In TN-C, Neutral and Earth share one wire. Severe harmonic server currents flow through the metal server racks, electrocuting engineers and corrupting network data packets!

B. Signal Reference Grid (SRG)

A **2-foot × 2-foot copper mesh** welded under the raised floor tiles. At high digital clock speeds (gigahertz), normal wires have high impedance. The SRG provides near-zero impedance ($Z \approx 0 \Omega$) to drain away electrostatic discharge (ESD) and radio-frequency noise.

Target Earth Resistance: $< 1.0 \text{ Ohm}$ (as per **IEEE 1100 Emerald Book**).



6. Solved University Numerical Problem (10-15 Marks)



The Given Exam Problem:

500 Server Racks @ 6 kW each. Target PUE = 1.40. 2N UPS Architecture (95% efficiency, 0.90 PF). Backup DG plant with N+1 redundancy using 1500 kVA generators (0.80 PF). 48 Hours diesel fuel runtime @ 0.25 L/kWh.

Step-by-Step Solved Arithmetic:

- IT Load:** $500 \times 6 \text{ kW} = \mathbf{3000 \text{ kW}} \text{ (3.0 MW)}$
- Total Facility Power:** $3000 \text{ kW} \times 1.40 \text{ (PUE)} = \mathbf{4200 \text{ kW}} \text{ (4.2 MW)}$
- UPS Capacity Sizing:** $\text{UPS Input (kW)} = \frac{3000}{0.95} = 3158 \text{ kW}$
 $\text{UPS kVA} = \frac{3158}{0.90 \text{ PF}} = \mathbf{3509 \text{ kVA}}$ Because it is a **2N architecture**:
 - Path A needs: $4 \times 1000 \text{ kVA}$ modules.
 - Path B needs: $4 \times 1000 \text{ kVA}$ modules.**Total = 8 modules of 1000 kVA each.**
- Diesel Generator Sizing (Must power full 4200 kW facility!):** $\text{Power per DG} = 1500 \text{ kVA} \times 0.80 = 1200 \text{ kW}$
 $N = \frac{4200 \text{ kW}}{1200 \text{ kW}} = 3.5 \rightarrow 4 \text{ Generators}$
 $\text{With N+1 Redundancy} = 4 + 1 = \mathbf{5 \text{ Generators of 1500 kVA each}}$
- Diesel Fuel Tank Size:** $\text{Total Energy} = 4200 \text{ kW} \times 48 \text{ Hours} = 201,600 \text{ kWh}$
 $\text{Net Fuel} = 201,600 \times 0.25 \text{ L/kWh} = 50,400 \text{ Litres}$
 $\text{Gross Tank Capacity (+10\% sludge bottom margin)} = \frac{50,400}{0.90} = \mathbf{56,000 \text{ Litres}}$



7. Ready-to-Write 15-Mark Exam Questions

Q1: Draw and explain the Single Line Diagram (SLD) of a Tier III Data Centre.

Checklist to draw: Two Utility Feeds —> Two RMUs —> Two Dry-Type Transformers (11kV to 415V) —> Main Low Voltage Switchboard with ATS —> Backup DG Plant —> Online Double Conversion UPS (Path A & Path B) —> Overhead Busway —> Dual Rack PDUs (A & B) —> Dual Power Supplies on Servers.

Q2: Explain earthing, TN-S scheme, and Signal Reference Grid.

Checklist to explain: Human safety + Digital noise immunity; TN-S separates Neutral and Protective Earth to stop harmonic currents; SRG 2ft×2ft copper mesh reduces high-frequency impedance to zero; IEEE 1100 < 1 Ohm standard.